

Yi Nathen Qing

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SUMMARY OF QUALIFICATIONS

- **Technical skills:** experience in Java, Python (NumPy, PyRosetta), MATLAB, CAD, Bash, and COMSOL.
- **Wet-lab skills:** DNA purification, E. coli protein purification, E. coli and mammalian cell culturing, colony PCR, GGA, two-photon microscopy, fluorescence spectroscopy, circular dichroism.

EDUCATION

University of Washington, School of Engineering | Seattle, WA

EXPECTED GRADUATION: June 20, 2027

- Major: Bioengineering: Data Science Option; Dean's List
- Related Coursework: BioE core, Matrix Algebra, Differential Equations, Databases, Algorithms, Fluids and Biotransport

PROFESSIONAL EXPERIENCE

Undergraduate Researcher | UW Baker Lab | Seattle, WA

Nov 2024 - Present

- Designing and testing de novo light harvesting proteins under the guidance of Avi Swartz as an undergraduate research intern.
- Currently designing protein nanowires capable of electron and energy transfer.
- Developed a custom computational protein design pipeline leveraging deep learning-based tools such as ProteinMPNN, RFDiffusion, and AlphaFold 3, combined with linear algebra-based parametric design.
- Conducting experimental validation through DNA cloning, bacteria culturing, protein expression, protein purification, and wet-lab assays such as fluorescence microscopy and circular dichroism.

Undergraduate Researcher | UW Institute of Protein Design JUPITER Cohort | Seattle, WA

Sep 2024 - Jun 2025

- Selected for the competitive Institute for Protein Design (IPD) JUPITER Cohort.
- Designed and tested phosphorylation-inducible heterodimers with Kathryn Shelley and Cullen Demakis using the protein dynamics pipeline Crispy-Shift and validated protein designs with ELISA assays.
- Presented research findings in Undergraduate Research Symposium poster session.

Undergraduate Researcher | UW Medicine Zheng Lab | Seattle, WA

Sep 2023 - Dec 2024

- Accepted as a first-year undergraduate to work with Dr. Sam Reyner in modeling microscopic functional intravascular topologies.
- Received training in two-photon microscopy and experience in organ-on-chip manufacturing.

Research Intern | Oregon Health and Science University Nakayama Lab | Portland, OR

Jun 2024 - Sep 2024

- Studied under the guidance of Dr. Karina Nakayama and members on biomaterial fabrication and statistical data analysis.
- Engineered open source Voron 3D printer into a bioprinter and gained experience in mammalian cell culture.
- Developed and preparing publications on methods for autonomous fabrication of shear-induced scaffolding material for tissue regeneration.

Student Mentor/Programmer | American Heritage School | Delray Beach, FL

Jul 2021 - Aug 2021

- Guided students on FTC and FRC robot programming and engineered chassis for software testing.
- Programmed software for HPCC competition project: autonomous security patrol robot with facial recognition and identification.

ADDITIONAL ACTIVITIES

Undergraduate Researcher | University of Washington | Seattle, WA

Sep 2023 - Present

- Selected for the University of Washington iGEM protein design and wet-lab team.
- 2023-2024 season worked on Gold medal winning Mighty Moeities project tackling bacteria antibiotic resistance. Helped ideate and experimentally validate protein linkers that extend cephalosporin lifetime by chemically linking it to human serum albumin.
- 2024-2025 season worked on Silver medal winning ExAway project helping immunocompromised patients against *P. aeruginosa*. Spearheaded the computational design of protein mini-binders that competitively inhibits bacterial toxin Exotoxin A.

Founding Member | Adam Biotech | Seattle, WA

Apr 2026 - Present

- Interviewed and selected for R&D team as founding member of student-led startup company manufacturing bio-printing equipment.
- Advanced to sweet 16 in UW Dempsey Business Competition and launching products in June.
- Currently operating in Biopunk biotechnology incubator in San Francisco.

High School Researcher | American Heritage School | Delray Beach, FL

Aug 2019 - May 2023

- Wrote a review paper on bottlenecks in in-vitro meat production under the guidance of Dr. Thompson.
- Programmed ML based software for classifying myosatellite tissue fitness through phenotypical features.

High School Robotics Competition Team | American Heritage School | Delray Beach, FL

Aug 2019 - May 2023

- Lead Programmer of high school FRC and FTC teams. Used Java and C++ extensively during machine vision development.
- Entered FRC Worlds quarterfinals during 2022-2023 school year.